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BEARING ASSEMBLY WITH INTERNAL PRIME-RETAINING PRESSURIZED LUBRICATION SYSTEM

Abstract

An internal pressurized lubrication system of a radial support bearing assembly includes oil passages within a bearing housing that comprise a charge-retaining section bounded by vertical or upward-sloping passages and configured to retain an oil charge therein during a shutdown. A vent bleeds air into the suction side of the passages during a shutdown, thereby preventing syphoning of the oil charge from the charge-retaining section. In embodiments, a rear cover plug is doubly sealed to the bearing housing body, and air is substantially unable to enter the oil passages via the bearing or via the vent during a shutdown, such that the passages remain oil-filled and the pump prime is retained for an extended time period. The bearing assembly can further include a thrust bearing, and/or an axial displacement measurement probe parallel with the shaft-driven oil pump. The radial support bearing can be a journal bearing.

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Background/Summary

FIELD OF THE INVENTION

[0001] The invention relates to bearing systems applicable to rotating shaft apparatus, and more particularly to bearing systems that comprise internal pressurized lubrication.

BACKGROUND OF THE INVENTION

[0002] Rotating shaft apparatus, including centrifugal pumps, motors, and turbines, rely on bearing systems to support a rotating shaft. In many cases, bearing systems require bearing lubrication provided by a liquid lubricant. Often, an oil is used as the lubricating liquid. However, it will be understood that the term “oil” is used herein generically to refer to any liquid lubricant, unless otherwise stated or required by context.

[0003] Various oil lubrication methods can be implemented in bearing systems, such as splash lubrication, oil mist lubrication, oil ring lubrication, and pressurized lubrication. Heat can be efficiently removed from bearings by circulating a lubrication oil within the bearing housing and cooling the lubrication oil in an oil cooler, which can be provided in the bearing housing. The heat can then be transferred to the bearing housing inner walls, which in turn conducts it to the exterior walls, ultimately allowing heat convection to the ambient air. Conversely, insufficient oil delivery to the bearings can result in insufficient oil film thickness and inadequate heat removal, which can increase the risk of direct contact between the bearing components, and ultimately to bearing failure, causing babbitt in journal bearings to deteriorate, and degrading the oil itself.

[0004] For large shaft sizes and/or rotational speeds, cost-effective oil lubrication methods, such as splash lubrication and oil ring lubrication, are often insufficient to deliver the required quantity of oil to lubricate bearings effectively, especially at large shaft diameters and rotational speeds. Instead, pressurized, “forced-fed” lubrication must be applied to the bearings in a sufficient quantity, and with adequate cooling. Typically, a shaft-driven lubrication pump fixed to the exterior of the bearing housing provides the required lubrication during normal operation, while a separate external oil pump is provided for startups, shutdowns, and loss of oil pressure situations. While reliable, this approach is complex, costly, and has a very large footprint. In addition, this approach often requires an external control system, which can delay pump startup after the required oil pressure is achieved. This can be disadvantageous if immediate startup is desired.

[0005] Another approach is to implement an internal oil pumping system within the bearing housing that includes an oil pump driven by the rotating shaft. It is critical for such internal lubrication systems to reliably deliver an adequate flow of oil to the bearings during normal operation, as well as during startups and shutdowns (transients). In particular, the oil that is provided in such systems tends to drain into an underlying “sump” when the shaft is not rotating, causing the pump to lose its “prime,” such that, during startup, the oil pump will pump air instead of oil. As a result, lubrication of the bearings is delayed during startup, which can increase bearing wear, and ultimately lead to bearing failure.

[0006] One approach is to partially submerge a lubricating rotor (rotating disk, impeller, etc) in the oil sump, so that it will immediately begin to pump oil once the shaft begins to rotate. However, this approach does not eliminate the initial lubrication delay that results from the draining of the oil from the oil passages that connect the lubricating rotor to the bearings. These passages must be

refilled with oil before lubrication of the bearings can begin. Also, submerging even a portion of the lubricating rotor in the oil sump tends to heat the oil in the sump when the pump is operating, thereby increasing the required oil cooling capacity of the system.

[0007] What is needed, therefore, is a bearing assembly comprising an internal, shaft-driven, pressurized lubrication system that minimizes or eliminates lubrication delays during system startup, without requiring direct contact between the lubricating rotor and the oil sump.

SUMMARY OF THE INVENTION

[0008] The present invention is a bearing assembly that is configured to support the shaft of a rotating shaft apparatus. The bearing assembly comprises at least one bearing that is lubricated by an internal, pressurized lubrication system comprising a lubricating liquid or “oil” that is pumped to the bearings from an “oil sump” by a lubricating rotor that is directly or indirectly driven by the shaft. The disclosed lubrication system minimizes or eliminates lubrication delays during system startup, without requiring direct contact between the lubricating rotor and the lubricating liquid in the oil sump. In embodiments, the lubricating liquid is an oil. However, it will be understood that the terms “oil” and “oil sump” are used herein generically to refer to any liquid lubricant and associated sump, unless otherwise stated or required by context.

[0009] The housing of the disclosed bearing assembly comprises a system of oil passages that maintains a quantity of oil, referred to herein as the “oil charge,” close to the bearings when the shaft is not rotating, even if there is a loss of prime on the suction side of the lubricating rotor, for example if rotation of the shaft remains halted during an extended period of time. When rotation of the shaft is restarted, this retained oil charge is immediately delivered to the bearings, even while the lubricating rotor is initially pumping an “air pocket” toward the bearings as it draws oil into the suction side and refills the remainder of the oil passages with oil.

[0010] The retained oil charge is adequate to provide initial lubrication of the bearings during startup, in embodiments during the first 10 to 15 shaft rotations, which is sufficient to sustain lubrication of the bearings as the air pocket reaches the bearings and passes through, for example during one to five shaft rotations, until the normal delivery of oil from the sump to the bearings has resumed. Accordingly, lubrication startup delay is virtually eliminated by the present invention.

[0011] The oil charge is contained within a charge-retaining section of the oil passages within the bearing housing. When the lubrication system is not operating, the retained oil charge is inhibited from draining into the sump because the charge-retaining section is bounded on both sides by segments of the oil passages that are either vertical, or slanted obliquely upward (from the perspective of the charge-retaining section). In addition, a vent is provided between the lubricating rotor and the charge-retaining section that is configured to allow air to enter the suction side during a system shutdown. The oil passages are tightly sealed, such that air is only able to enter the oil passages during a system shutdown via the vent, and possibly also via the bearings. In embodiments, the vent is configured such that the rate at which air can enter the suction side of the oil passages during a shutdown via the vent is slightly faster than the rate at which air is able to enter the oil passages via the bearings, thereby preventing the retained oil charge from being syphoned away as the remainder of the oil in the oil passages drains into the sump. In embodiments, the vent also functions as a flow restrictor configured to control a flow of the oil from the lubricating rotor to the bearings.

[0012] If air is substantially unable to enter the oil passages through the bearings during a shutdown, for example if the bearings are journal bearings, then the vent can be configured to allow only a very slow entry of air, such that little if any oil drains from the oil passages into the sump, and the pumping system retains its prime, unless the shutdown continues for an extended period of time.

[0013] In embodiments, the disclosed bearing assembly includes a thrust bearing in addition to one or more radial support bearings. Some embodiments include an axial displacement measurement probe located in parallel with the shaft-driven lubricating rotor. And in various embodiments the

disclosed bearing housing is a first of a plurality of interconnected bearing housings, wherein the lubricating rotor of the first bearing housing is configured to circulate pressurized oil to all of the bearing housings. In some of these embodiments the oil passages of each of the bearing housings include charge-retaining sections, each of which is configured to retain an oil charge during a shutdown.

[0014] The present invention is a bearing assembly that includes a bearing housing, a support bearing installed in the bearing housing and configured to radially support a horizontal, rotatable shaft, and a lubrication system. The lubrication system includes a sump formed in a lower section of the bearing housing, the sump being configured to contain a lubricating liquid, a plurality of lubricating liquid passages formed in the bearing housing, and a lubricating rotor installed in the bearing housing above the sump and configured to be rotated by the shaft.

[0015] The lubricating rotor, in combination with the lubricating liquid passages, is configured to draw the lubricating liquid from the sump, pressurize the lubricating liquid, and direct the pressurized lubricating liquid to the support bearing. The lubricating liquid passages comprise a suction passage configured to direct the lubricating liquid at ambient pressure from the sump to the lubricating rotor, a vertically downward or downward-sloping passage configured to receive the pressurized lubricating liquid from the lubricating rotor, and a charge-retaining section extending substantially horizontally from the vertically downward or downward-sloping passage to a lubrication delivery passage proximate the support bearing, the lubrication delivery passage providing liquid access from the charge-retaining section to the support bearing via a vertically upward or upwardly sloping segment, the charge-retaining section being thereby configured to retain a charge of the lubricating liquid when the lubricating rotor is not rotating.

[0016] Embodiments further include a vent configured to control a flow of air into the suction passage when the lubricating rotor is not rotating, thereby preventing the charge of the lubricating liquid from being syphoned out of the charge-retaining section as the lubricating liquid is gravitationally drawn from the suction passage to the sump. In some of these embodiments, the vent also functions as a flow restrictor that is configured to control a flow of the pressurized lubricating liquid through the vertically downward or downward-sloping passage when the lubricating rotor is rotating. In any of these embodiments, the vent can be located in the vertically downward or downward-sloping passage.

[0017] In any of the above embodiments, the bearing housing can include a housing body and a rear cover; at least one of the lubricating liquid passages can be formed in the rear cover, such that the lubricating liquid flows between the rear cover and the housing body when the lubricating rotor is rotating; and leakage of air between the rear cover and the housing body, and thereby into the lubricating liquid passages, can be prevented by a first seal formed between the rear cover and the housing body axially distal to the lubricating liquid passages within the rear cover, in combination with a second seal formed between the rear cover and the housing body axially proximal to the lubricating liquid passages within the rear cover.

[0018] In any of the above embodiments, when the lubricating rotor is not rotating, air can be substantially unable to enter the lubricating liquid passages, thereby preventing the lubricating liquid from flowing gravitationally to the sump when the lubricating rotor is not rotating.

[0019] In any of the above embodiments, the support bearing can be a journal bearing.

[0020] Any of the above embodiments can further include a thrust bearing configured to limit axial movements of the rotating shaft. In some of these embodiments, the thrust bearing is lubricated by the lubrication system. Any of these embodiments can further include an axial displacement measurement probe cooperative with the thrust bearing and located proximate the lubricating rotor.

[0021] In any of the above embodiments, the vertically upward or upwardly sloping segment of the lubrication delivery passage can be a segment of a lubrication collar that surrounds the support bearing, the lubrication collar being in liquid communication with the support bearing via lubrication inlets located above the charge-retaining section of the lubricating liquid passages.

[0022] And in any of the above embodiments, during a rotation startup of the shaft, the charge of the lubricating liquid can be adequate to provide lubrication of the bearings at least during the first ten rotations of the shaft.

[0023] The features and advantages described herein are not all-inclusive and, in particular, many additional features and advantages will be apparent to one of ordinary skill in the art in view of the drawings, specification, and claims. Moreover, it should be noted that the language used in the specification has been principally selected for readability and instructional purposes, and not to limit the scope of the inventive subject matter.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0024] FIG. 1 is a perspective view, drawn to scale, of an exemplary embodiment of the present invention;

[0025] FIG. 2 is a longitudinal sectional view, drawn to scale, of the exemplary embodiment;

[0026] FIG. 3 is a lateral sectional view, drawn to scale, of the exemplary embodiment that intersects the rear plate of the bearing assembly;

[0027] FIG. 4 is a horizontal sectional view, drawn to scale, of the exemplary embodiment;

[0028] FIG. 5A is a lateral sectional view, drawn to scale, of the exemplary embodiment that intersects the lubrication inlet port of the bearing assembly;

[0029] FIG. 5B is a closeup view, drawn to scale, of a portion of the sectional view of FIG. 5A; and

[0030] FIG. 6 is a simplified illustration of the oil passages that direct oil from the lubricating rotor to the lubrication port.

DETAILED DESCRIPTION

[0031] The present invention is a bearing assembly that is configured to support the shaft of a rotating shaft apparatus. The bearing assembly comprises at least one bearing that is lubricated by an internal, pressurized lubrication system comprising a lubricating liquid or “oil” that is pumped to the bearings from an “oil sump” by a lubricating rotor that is directly or indirectly driven by the shaft. The disclosed lubrication system minimizes or eliminates lubrication delays during system startup, without requiring direct contact between the lubricating rotor and the lubricating liquid in the oil sump. In embodiments, the lubricating liquid is an oil. However, it will be understood that the terms “oil” and “oil sump” are used herein generically to refer to any liquid lubricant and associated sump, unless otherwise stated or required by context.

[0032] A perspective view of an exemplary embodiment **100** of the present invention is presented in FIG. 1. FIGS. 2-5B are sectional views of this exemplary embodiment having section planes as indicated in FIG. 1.

[0033] With reference to FIG. 2, the exemplary embodiment comprises a journal bearing **200** that radially supports a rotating shaft (not shown), as well as a thrust bearing **202** that limits axial movements of the shaft. The bearing housing **100** of the exemplary embodiment comprises a lubrication system that includes an oil sump **204** and a system of oil passages, including a “suction side” passage **208** having a suction inlet **220** through which oil is drawn from the sump **204** by the lubricating rotor **210**.

[0034] A set of cooling coils **206** is provided in the sump **204** through which an oil cooling fluid is circulated. A rear “cover” or plug **216** comprises a suction side extension **218** through which the oil is transferred from the suction side passage **208** to the lubricating rotor **210**. The rear cover **216** is sealed to the body **214** of the bearing housing by a pair of O-ring seals **212** that surround the rear cover **216**. In combination, these O-ring seals **212** prevent air from leaking from either side of the rear cover **216** into any of the oil passages that extend from the body **214** into the rear cover **216**. In similar embodiments, air is prevented from leaking into any of the oil passages by individual O-

rings associated with each of the oil passage junctions, or by other sealing mechanisms as are known in the art.

[0035] When the shaft is not rotating, the oil passages maintain a quantity of oil, referred to herein as an “oil charge,” in a “charge-retaining section” (**600** in FIG. **6**) that is close to the bearings, even if there is a loss of prime on the suction side **208, 218** of the lubricating rotor **210**, for example if rotation of the shaft remains halted during an extended period of time.

[0036] When rotation of the shaft is restarted, the retained oil charge is immediately delivered to the bearings **200**, even while the lubricating rotor **210** is initially pumping an “air pocket” toward the bearings **200** as it draws oil from the sump **204** and refills the remainder of the oil passages **208, 218** with oil. The retained oil charge is adequate to provide initial lubrication of the bearings **200** during startup, in embodiments during the first 10 to 15 shaft rotations, which is sufficient to sustain lubrication of the bearings **200** as the air pocket reaches the bearings and passes through, for example during one to five shaft rotations, until the normal delivery of oil from the sump **204** to the bearings **200** has resumed. Accordingly, lubrication startup delay is virtually eliminated by the present invention.

[0037] With reference to FIG. **3** and FIG. **6**, in the exemplary embodiment, oil from the suction side extension **218** is directed through a rotor inlet **300** to the lubricating rotor **210**, and pressurized oil is returned to a rotor outlet **302**, from which it flows through a downward sloping passage **304** to a first segment **306** of the charge retaining section **600**, and then into a second, horizontal segment **308** of the charge retaining section **600**. Note that passage **304** is sloped “downward” relative to the lubricating rotor **210**, but “upward” relative to the charge retaining section **600**. A flow restrictor **310** that acts as a vent during a system shutdown is provided between the lubricating rotor **210** and the charge-retaining section **600**. In the exemplary embodiment, an axial displacement measurement probe **312** is located in parallel with the shaft driven lubricating rotor **210**. In similar embodiments, the axial displacement measurement probe **312** is provided at other locations relative to the lubricating rotor **210**.

[0038] With reference to FIG. **4** and FIG. **6**, the second segment **308** of the charge-retaining section **600** continues within the bearing housing **100** parallel to the shaft to a bearing lubrication inlet port **400**, which extends laterally inward to a lubrication collar **402**. In the exemplary embodiment, the bearing housing **100** is configured to enable delivery of pressurized oil to additional bearing housings via an oil circulation port **404** that is cooperative with the flow restrictor **310**. In some of these embodiments, charge-retaining sections are included in the oil passages of each of the bearing housings that are configured to retain oil charges during a shutdown.

[0039] With reference to FIGS. **5A, 5B**, and **6**, the oil flows from the lubrication inlet port **400** into the lubrication collar **402**, and then through lubrication inlets **502** to the axial support bearing **200**, which in the exemplary embodiment is a journal bearing **200**. The flow path **504** of the oil is indicated in the enlargement of FIG. **5B**. It can be seen in the figure that the oil is required to flow almost vertically in the lubrication collar **402** before it can enter the lubrication inlet **502** and flow to the bearing **200**. The lubrication collar **402** thereby terminates the charge-retaining section **600** of the oil passages.

[0040] FIG. **6** is a simplified illustration of the path that is followed by lubricating oil in the exemplary embodiment from the lubricating rotor **210** to the bearings **200**. For simplicity of illustration, the passage segments are indicated as if they were co-planar. It can be seen in the illustration that the charge-retaining section **600** includes passages **306, 308**, and **400**, and is bounded by the upward sloping passage **304** (relative to the charge-retaining section **600**) and the nearly vertical passage **402**.

[0041] The oil passages are tightly sealed, including by the seals **212** between the rear cover **216** and the body **214** of the bearing housing **100**, such that air is only able to enter the oil passages during a system shutdown via the flow restrictor **310**, and possibly also via the bearings **200**. In embodiments, the flow restrictor **310** is configured such that the rate at which air can enter the oil

passages during a shutdown via the flow restrictor **310** is slightly faster than the rate at which air is able to enter the oil passages via the bearings **200**, thereby preventing the retained oil charge from being syphoned away as the oil from the remainder of the oil passages drains into the sump **204**. [0042] In the exemplary embodiment, the bearings **200** are journal bearings **100**, which substantially prevent air from entering the oil passages via the bearings **200** during a shutdown. Accordingly, the flow restrictor **310** in the exemplary embodiment is configured to allow only a very slow entry of air, such that little if any oil drains from the oil passages into the sump **204** during a shutdown, all of the oil passages remain filled with oil, and the pumping system retains its prime, unless the shutdown continues for an extended period of time.

[0043] And even if the prime is lost during a shutdown of the exemplary embodiment, the retained oil charge is inhibited from draining into the sump **204** because the charge-retaining section **600** is bounded on both sides by segments of the oil passages that are either vertical **402**, or slanted obliquely upward **304**, and because the flow restrictor **310** prevents the oil in the charge-retaining section **600** from being syphoned away.

[0044] The foregoing description of the embodiments of the invention has been presented for the purposes of illustration and description. Each and every page of this submission, and all contents thereon, however characterized, identified, or numbered, is considered a substantive part of this application for all purposes, irrespective of form or placement within the application. This specification is not intended to be exhaustive or to limit the invention to the precise form disclosed. Many modifications and variations are possible in light of this disclosure.

[0045] Although the present application is shown in a limited number of forms, the scope of the disclosure is not limited to just these forms, but is amenable to various changes and modifications. The present application does not explicitly recite all possible combinations of features that fall within the scope of the disclosure. The features disclosed herein for the various embodiments can generally be interchanged and combined into any combinations that are not self-contradictory without departing from the scope of the disclosure. In particular, the limitations presented in dependent claims below can be combined with their corresponding independent claims in any number and in any order without departing from the scope of this disclosure, unless the dependent claims are logically incompatible with each other.

Claims

1. A bearing assembly comprising: a bearing housing; a support bearing installed in the bearing housing and configured to radially support a horizontal, rotatable shaft; and a lubrication system comprising: a sump formed in a lower section of the bearing housing, the sump being configured to contain a lubricating liquid; plurality of lubricating liquid passages formed in the bearing housing; and a lubricating rotor installed in the bearing housing above the sump and configured to be rotated by the shaft, the lubricating rotor, in combination with the lubricating liquid passages, being configured to draw the lubricating liquid from the sump, pressurize the lubricating liquid, and direct the pressurized lubricating liquid to the support bearing; wherein the lubricating liquid passages comprise: a suction passage configured to direct the lubricating liquid at ambient pressure from the sump to the lubricating rotor; a vertically downward or downward-sloping passage configured to receive the pressurized lubricating liquid from the lubricating rotor; and a charge-retaining section extending substantially horizontally from the vertically downward or downward-sloping passage to a lubrication delivery passage proximate the support bearing, the lubrication delivery passage providing liquid access from the charge-retaining section to the support bearing via a vertically upward or upwardly sloping segment, the charge-retaining section being thereby configured to retain a charge of the lubricating liquid when the lubricating rotor is not rotating.
2. The bearing assembly of claim 1, further comprising a vent configured to control a flow of air into the suction passage when the lubricating rotor is not rotating, thereby preventing the charge of

the lubricating liquid from being syphoned out of the charge-retaining section as the lubricating liquid is gravitationally drawn from the suction passage to the sump.

3. The bearing assembly of claim 2, wherein the vent also functions as a flow restrictor that is configured to control a flow of the pressurized lubricating liquid through the vertically downward or downward-sloping passage when the lubricating rotor is rotating.
 4. The bearing assembly of claim 2, wherein the vent is located in the vertically downward or downward-sloping passage.
 5. The bearing assembly of claim 1, wherein: the bearing housing comprises a housing body and a rear cover; at least one of the lubricating liquid passages is formed in the rear cover, such that the lubricating liquid flows between the rear cover and the housing body when the lubricating rotor is rotating; and leakage of air between the rear cover and the housing body, and thereby into the lubricating liquid passages, is prevented by a first seal formed between the rear cover and the housing body axially distal to the lubricating liquid passages within the rear cover, in combination with a second seal formed between the rear cover and the housing body axially proximal to the lubricating liquid passages within the rear cover.
 6. The bearing assembly of claim 1, wherein when the lubricating rotor is not rotating, air is substantially unable to enter the lubricating liquid passages, thereby preventing the lubricating liquid from flowing gravitationally to the sump when the lubricating rotor is not rotating.
 7. The bearing assembly of claim 1, wherein the support bearing is a journal bearing.
 8. The bearing assembly of claim 1, further comprising a thrust bearing configured to limit axial movements of the rotating shaft.
 9. The bearing assembly of claim 8, wherein the thrust bearing is lubricated by the lubrication system.
 10. The bearing assembly of claim 8, further comprising an axial displacement measurement probe cooperative with the thrust bearing and located proximate the lubricating rotor.
 11. The bearing assembly of claim 1, wherein the vertically upward or upwardly sloping segment of the lubrication delivery passage is a segment of a lubrication collar that surrounds the support bearing, the lubrication collar being in liquid communication with the support bearing via lubrication inlets located above the charge-retaining section of the lubricating liquid passages.
 12. The bearing assembly of claim 1, wherein, during a rotation startup of the shaft, the charge of the lubricating liquid is adequate to provide lubrication of the bearings at least during the first ten rotations of the shaft.
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